

MAT 201-001, Calculus and Analytic Geometry 3, Spring 2024**8 - 9:45 am, Mondays and Wednesdays****Dr. Mary Kassebaum, mokassebaum@aacc.edu**

Extend the ideas of limits, differentiation and integration to two or more dimensions. Topics include vectors and vector functions, partial differentiation, gradients, directional derivatives and tangent planes, optimization, Lagrange multipliers, iterated integration, line and surface integrals, and the classical theorems of Green, Gauss and Stokes.

At the conclusion of this course, students will be able to

- a. Analyze curves and surfaces using partial differentiation, directional derivatives, and the gradient operator.*
- b. Analyze space curves, including their curvature and arclength, as well as the velocity and acceleration of an object on a curve.*
- c. Solve optimization problems in several variables.*
- d. Extend integration to multivariate functions for applications including finding areas, volumes, and moments.*
- e. Apply the curl and divergence operators to vector fields.*
- f. Calculate the circulation of a vector field around the boundary of a plane region with and without Green's Theorem.*
- g. Calculate the circulation of a vector field around the boundary of a surface with and without Stokes' Theorem.*
- h. Calculate the flux of a vector field through a closed surface with and without the Divergence (Gauss') Theorem.*

Meet with Dr. Kassebaum:

In class: Mondays and Wednesdays at 8 am. Class meetings are required—not optional. If you miss class, contact a classmate or Dr. Kassebaum to find out what you missed.

Homework sessions: Monday: 12-1:30 pm (in person) **See Meetings page (Canvas) for locations/links!**

Wednesday: 12-1:30 pm (in person)

There will be other times, too...please respond to the Canvas Homework Session Survey!

...and other times by request—please email through Canvas to set up a time.

Email Dr. Kassebaum: Please send ALL e-mail through Canvas (preferred) or directly to me at mokassebaum@aacc.edu. I try to respond to e-mail within one business day. *Please do not email or message me through WebAssign—I do not see these messages.*

Materials required for this course, this semester:

- **WebAssign Code OR paper text:** Calculus: Early Transcendentals, 9th edition, or Multivariable Calculus, 9th edition, by James Stewart, published by CENGAGE. WebAssign course/e-book is available for this section—see our Canvas course for our class key and details.
- **Graphing calculator:** TI-83 or TI-84 highly recommended; **other calculators require advance approval** from Dr. Kassebaum for use on exams.
- **Scanning app or scanner** for scanning notes to pdf format to upload for grading.

If you need to obtain some technology (computer/laptop/Chromebook, webcam), the following sources may help:

1. The AACC Library has some Webcams and laptops to rent for **FREE**. To request, email the library at circdesk@aacc.edu from your myAACC email. In the email, give your full name, course schedule, semester, and the item(s) you are requesting to rent. Check your myAACC email regularly for follow-up directions from the Library. ***There are a limited number of items; contact them as soon as possible with your request.***
2. If you are receiving Financial Aid, funds can often be used to purchase or rent some items from the AACC Bookstore. Contact the bookstore at websales@aacc.edu, or by calling 410-777-2082.

Important facts, dates, and notices:

- **Exams** will be taken during the first part of class time on **Wednesday, February 21; Wednesday, March 13; and Monday, April 15. The final exam is scheduled for Monday, May 13, 8:00 – 10:00 am, in the regular classroom.**
- Refund and audit deadlines are posted on MyAACC -> My Class Schedule. The last day to withdraw from this course is April 22, 2024.

Assignments, exams, and grading for this course:

- There will be a **Course Resource assignment submitted IN CLASS** worth 24 points. There will be **graded “warm-up” assignments** which will be **submitted online** and graded about once per week throughout the semester (graded out of 50 points). These assignments are given in the notes. ***Since there will be an extra assignment to use as make-up/extra credit, Warm-Up assignments are not accepted after the due date and time.***
- There will be **5 graded exam review assignments** worth 30 points each for a total of 150 points. Original copies of this assignment will be **submitted IN CLASS**. (Please take pictures or scans of your work prior to turning in; assignments will not be graded prior to exam, but assignment key will be posted for studying.) If turned in (in office MATH 231) after the deadline but within 48 hours, a graded review assignment is worth (at most) half credit. ***No other extensions or make-ups will be granted.***
- There will be 11 **online homework discussion** assignments worth 10 points each, graded out of 100 points. ***Late discussion assignments are not typically accepted, but you should ask if you have special circumstances.***
- There will be three 100-point exams given (note dates given previously). There will also be a 200-point **comprehensive final exam given on Monday, May 13. If you miss an exam, email me (Canvas or mokassebaum@aacc.edu) as soon as possible!**
- **May 1-7 is “makeup time” for MAT 201-001.** If you would like to retake any **one** of the three exams given in the course, you may do so during this time in the Testing Center, SUN 240. Instructions given after exam 3.
- **Other assignments** may be given during the semester if deemed necessary. Points earned will be included in the point total (they are **not** “extra credit”).
- **Extra credit** assignments (in addition to extra points already built into the course) may be offered occasionally. They will always be announced in class or via CANVAS, clearly identified as “extra,” and will be open to all students in the course.
- **How grades will be assigned:** Warm-up assignments, Homework Session Survey, and the Course Resource assignment (75 points), exam review assignments (150 points), on-line discussions (100 points), exams (300 points), and the final exam (200 points) give a total of 825 points possible in the course (plus any other graded assignments necessary). You will receive an A for 90% or more of the total points, at least a B for 80% or more, at least a C for 70% or more, at least a D for 60% or more.

Dr. Kassebaum’s expectations of her students:

- **You are responsible for material covered in class sessions, posted on Canvas, or emailed.**
- **You are expected to use the CANVAS course site extensively** to guide your reading of the text, check course e-mail, check your grade, and access course materials.
- **You are expected to be courteous to your fellow students and to your professor.** We all have responsibilities and challenges in our lives. ***We will be kind and supportive of each other this semester.***
- **You are expected to put away your cellphone and all other electronic devices while inside the classroom during class time.** If you would like to use an iPad or other tablet for notetaking, please get permission before you do this. Failure to abide by this policy will trigger a knowledge demonstration opportunity for the entire class; this new assignment will be added to the point total for the course and will not be extra credit.
- This is a 4-credit course delivered in 15 weeks. **This requires that you spend at least 12-16 hours each week completing the work for this class.** This includes class attendance as well as your notetaking, text/ebook-reading, and problem-solving time. Please set this time aside today.
- **Do not wait to get help if you are confused!** Each piece of new material builds on what has been presented before. USE the homework sessions, e-mail me, JOIN the Virtual Math Tutoring Lab (click the link on our Canvas home page!), USE the in-person tutoring at LIB 102, find or create a study group.
- While AACC does not require masking or vaccines, as members of a college community, **you are expected to take reasonable precautions to prevent transmissible illnesses** like COVID-19. You are encouraged to stay home and seek treatment if you are ill. Please contact me to arrange completion of your coursework.

Recommended activities for learning the material of each section:

- **Attend our class meetings!** This is an opportunity to see the derivation of theorems and to watch problem solutions unfold. If you must miss a class, please use the notes from the class posted on Canvas, and please contact a classmate or Dr. Kassebaum to find out what else happened.
- **Participate in our class meetings!** *There are few things more dull than watching someone write!* Make it interesting by engaging in the material—**BEFORE, DURING, AND AFTER CLASS...**
 - **BEFORE:** Look at your notes from the previous class day and the (blank) notes for the next class; fill in the required “warm-up” pages, using your text/e-book; complete at least 3 exercises from assigned homework sections (from previous class).
 - **DURING:** Ask questions on homework before and after class; take notes during class. **CAUTION: Dr. Kassebaum’s notes may include some pre-drawn figures or pre-written definitions and theorems.** Concentrate on understanding and writing what she is covering. **You will be able to go back later and fill in the blanks in your notes** from the posted class notes on Canvas.
 - **AFTER:** Re-work the examples covered in the notes for that day; use the notes posted (after class) on Canvas to fill in the note pages you did not finish during class; start at least 3 exercises from each homework section covered in today’s class. **If you don’t know how to start the homework, go to today’s homework session.**
- **Work (and re-work) your discussion homework!** You can’t learn a skill solely by watching someone else do it. You have to get in there and try it yourself. **You will not master the material if you only do the few homework exercises you must post to discussions!** By exam time, you should have worked and practiced each discussion and exam review problem several times.
- **Talk to your classmates, friends, and Dr. Kassebaum about math!** Verbalizing the mathematics you are learning creates new connections and understanding in your brain.

Learner Support Services: If you or any other student you know faces challenges securing their food, transportation, housing, or course materials and believes this may affect their performance in this course, you are urged to contact the Office of the Dean of Student Engagement for assistance. Please email studenthelp@uakron.edu or call 410-777-2511.

Please understand...

- I care about you, and I want you to be successful.
- I enjoy working with you, and I want you to see yourself as a mathematician and a future STEM professional.

**If there is anything this semester
—illness, work schedule, or even the course policies and schedule—
that is a barrier to your success, please contact me.**

**It is my policy that “it never hurts to ask.”
Please feel free to ask for what you need; if I can figure out how to provide it
without being unfair to others, I will.
The easiest way to ask is to send me an email through Canvas, or directly to
mokassebaum@uakron.edu.**

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Approximate Assignment Schedule; changes will be announced in class and/or posted on Canvas.

Week: Class meetings	In-class topics	Assignments
W1: 1/22, 24	Course introduction, 12.1 – 12.3	Warm-Up 1 due online by 8 am, 1/24 Course Resource Assignment, due 11:59 pm, 1/24
W2: 1/29, 31	12.4 – 12.6	Warm-Up 2 due online by 8 am, 1/29 Discussion 1: post by 2/4; comment by 2/7
W3: 2/5, 7	12.6, 13.1-13.2 Graded Review for Exam 1—part 1	Warm-Up 3 due online by 8 am, 2/5 Discussion 2: post by 2/10; comment by 2/13 Graded Review for Exam 1—part 1 due in class 2/7 (negotiable).
W4: 2/12, 14	13.2 – 13.4	Warm-Up 4 due online by 8 am, 2/12 Discussion 3: post by 2/17; comment by 2/20
W5: 2/19, 21	Graded Review for Exam 1 – part 2 Exam 1 (2/21) 14.1 – 14.2	Graded Review for Exam 1 – part 2 due in class, 2/19 (not as negotiable). Exam 1 during first part of class 2/21
W6: 2/26, 2/28	14.2 – 14.4	Warm-Up 5 due online by 8 am, 2/26 Discussion 4: post by 3/2; comment by 3/5
W7: 3/4, 6	14.4 – 14.8	Warm-Up 6 due online by 8 am, 3/4 Discussion 5: post by 3/9; comment by 3/12
W8: 3/11, 13	Graded Review for Exam 2 Exam 2 (3/13) 15.1	Graded Review for Exam 2 due in class, 3/11 (not very negotiable). Exam 2 during first part of class 3/13.
AACC Spring Break	No classes 3/18-24	Campus offices closed
W9: 3/25, 27	15.2 – 15.4	Warm-Up 7 due online by 8 am, 3/25 Discussion 6: post by 3/30; comment by 4/2
W10: 4/1, 3	15.5 – 15.8	Warm-Up 8 due online by 8 am, 4/1 Discussion 7: post by 4/6; comment by 4/9
W11: 4/8, 10	15.9 Graded Review for Exam 3	Warm-Up 9 due online by 8 am, 4/8 Discussion 8: post by 4/13; comment by 4/16 Graded Review for Exam 3 due in class 4/10 (negotiable).
W12: 4/15, 17	Exam 3 (4/15) 16.1 – 16.3	Exam 3 during first part of class, 4/15. Warm-Up 10 due online by 8 am, 4/17
W13: 4/22, 24	16.3 – 16.7	Warm-Up 11 due online by 8 am, 4/22 Discussion 9: post by 4/27; comment by 4/30
W14: 4/29, 5/1	16.7 – 16.10 Make-up exam 5/1-7	Warm-Up 12 due online by 8 am, 4/29 Discussion 10: post by 5/4; comment by 5/7 Make-up exam 5/1-7 in SUN 240
W15: 5/6, 8	16.10 Graded Review of Module 4 Make-up exam 5/1-7	Discussion 11: post by 5/9 (THURSDAY!); comment by 5/11 (SATURDAY!) Graded Review of Module 4 due in class 5/8 (slightly negotiable). Make-up exam 5/1-7 in SUN 240
Finals Week 5/13-17	Final Exam , Monday, May 13, 8-10am, in regular classroom	All course work other than the Final Exam will be submitted prior to 8 am, May 13, 2024.

College policies for all AACC students:

Emergency College Closure For closure information, students can check www.aacc.edu, and can sign up for text message alerts using the [Stay Informed with Campus Alerts](#) web page. If class is cancelled because of an emergency college closure, please check your course and email in Canvas for possible announcements.

Academic Integrity Policy Anne Arundel Community College, with a central mission of producing learning and a belief that individuals be given the opportunity to fully develop their potential, is committed to upholding rigorous and fair standards of student learning and achievement. Achieving successful student learning is dependent upon a dedication to academic integrity on the part of all members of the college community. Without academic integrity, students gain unfair advantage over others and impede their own development.

In support of this aim, Anne Arundel Community College requires all students to exhibit academic integrity in all their academic work. A culture of academic integrity, a unifying principle at AACC and all academic communities, is built upon respect for others' work, commitment to doing one's own work, and intolerance for academic dishonesty in all its forms. Students are responsible for reviewing and complying with the College's Academic Integrity Policy. A student found in violation of this policy shall be subject to appropriate sanctions, up to and including expulsion from the College, in accordance with the Resolving Allegations of Academic Misconduct Procedure and the Academic Misconduct Hearing Procedure. For more information, visit the [Academic Integrity Policy](#) and [AACC Policies webpage for the Academic Integrity Procedures](#) (.).

Student Conduct Policy Students shall always conduct themselves in a manner that demonstrates mutual respect and courtesy, displays appropriate standards of behavior, and refrains from any actions or in actions that impinge on the rights of others or disrupt the teaching and/or learning process or the operations of the college. A student found in violation of this policy or any other College policy shall be subject to appropriate sanctions, up to and including expulsion from the College, in accordance with the student conduct procedures. The full text of the policy is available on the college's [AACC Policies webpage](#) and in the Student Handbook and College catalog. Additional information for [Code of Conduct webpage](#).

Acceptable Use of Information Technology Resources Policy This policy governs the acceptable use of the college information resources by anyone. This policy applies to students enrolled in this course at any time they are using college resources. The goal of the usage policy is to encourage an environment of learning in which all students can interact in an open, legal, and ethical manner. The full text of the policy is available on the AACC [Acceptable Use of Technology Resources Policy webpage](#) that implement the policy are available on the AACC [Acceptable Use of Information Technology Resources Procedures](#).

Notice of Nondiscrimination updated June 1, 2023 The college prohibits all types of discrimination including, but not limited to, race – including hair texture and hairstyle – color, religion or creed, ancestry or national origin, sex, age, marital status, physical or mental disability, sexual orientation, gender identity, veteran status, citizenship, and genetic information. Complaints of discrimination shall be reported to the college's chief compliance and fair practices officer. For information regarding the college's compliance and complaints concerning discrimination, contact the chief compliance and fair practices officer and Title IX coordinator at complianceofficer@aacc.edu or 410-777-1239, or call Maryland Relay at 711.

Accommodations Students who need a reasonable accommodation should contact [Disability Support Services](#) at DSS@aacc.edu or 410-777-1411.

Religious Accommodations Policy Anne Arundel Community College prohibits discrimination, harassment, and retaliation on the basis of religion. The purpose of this Religious Accommodations Policy is to describe the rights of Employees and Students to request and receive Religious Accommodations in accordance with the College's policies and relevant laws and regulations regarding Religious Beliefs. For more information, review the [Religious Accommodations Policy webpage](#). Also, review the [Student Religious Accommodations Procedure webpage](#).

Free Support Services Available on Campus for AACC Students

Service	Location	Phone
Counseling Services	Student Union (SUN 203); appointment request via https://counseling.aacc.edu/TitaniumWeb/	410-777-7111 or email counseling@aacc.edu
Military/Veterans Resource Center	Truxal Library, room 112 https://www.aacc.edu/military/	410-777-4838 or email veterans@aacc.edu
Disability Support Services	Student Union (SUN 140), https://www.aacc.edu/resources/disability-support-services/	410-777-1411
Student Achievement and Success Program (SASP)	https://www.aacc.edu/about/administrative-offices/student-achievement-and-success-program/	410-777-2530
Substance Use Prevention, Education, and Resources	https://www.aacc.edu/resources/health-and-personal-counseling/substance-abuse-education/#d.en.5787	410-777-2480 (Shawn Grim, RN, Health & Wellness Center)
Writing Center	Truxal Library, room 136, as well as virtual sessions https://www.aacc.edu/resources/academic-services/tutoring/writing-center/	Email jgrabin@aacc.edu (Dr. Jessica Rabin, coordinator)
Career Services	https://www.aacc.edu/resources/careers-and-employment/career-services/	Email jbsturtevant@aacc.edu
Student Engagement	Student Union (SUN 225) https://www.aacc.edu/about/administrative-offices/student-engagement/	410-777-2218
Student Food Pantry	Student Union (SUN 219) Glen Burnie Town Center (GBTC 311) Food pantry visits by appointment at AACC HelpLink appointment form	410-777-2218
HelpLink	Student Union (SUN 210) Glen Burnie Town Center (GBTC 311) https://www.aacc.edu/resources/health-and-personal-counseling/emergency-support/helplink-grant/	410-777-2511, or email studenthelplink@aacc.edu
AACC Rainbow Network	Look for Rainbow Network (ARN) signs around campus that identify offices of faculty and staff who work to end discrimination against individuals identifying as LGBTQ+ throughout our campus community and the community at large. https://www.aacc.edu/about/diversity-and-inclusivity/rainbow-network/	
Any student facing challenges securing their food, transportation, housing, or course materials is urged to contact the Office of Student Engagement for support. Call 410-777-2218. https://www.aacc.edu/about/administrative-offices/student-engagement/		